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Lab 1 Preparation:

Thyristor:

1. What can you tell me about the maximum efficiency of this circuit

$$\text{efficiency} = \frac{\text{power delivered to load}}{\text{power supplied by the source}}$$

$$\text{efficiency} = \frac{I^2 R_L}{I^2 (R_L + R_s)}$$

If we want the efficiency to be maximum, then we need a large term in the numerator and a small term in the denominator.

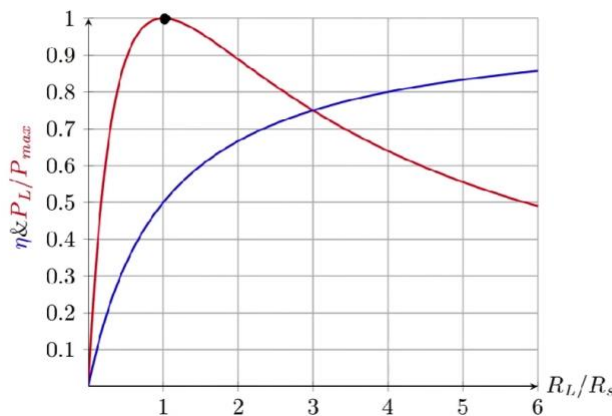
The efficiency approaches 100% as the source impedance gets small or as the load impedance gets large.

- ➔ In our circuit we can see from the schematic that there is no source impedance which means that the efficiency is 100% and we have a maximum efficiency

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2. What can you tell me about the maximum power delivered to the load?

In case R_s is not ideal, The graph below shows the relationship between the maximum power and the efficiency.

Maximum Power Transfer



The maximum power delivered to the load depends on the voltage applied to it.

The light bulb is rated at 12V and 10 W.

- ➔ Since R_s is ideal, The **maximum power** delivered to the light bulb will be **10W**, which occurs when the circuit is allowing the full 12V RMS AC voltage to reach the bulb without phase delay. Any dimming via the phase control will reduce the power delivered to the bulb below this maximum value.

3. Can we turn the SCR off?

It's possible to manually turn off the SCR in our circuit but it requires intervention to force the SCR to stop conducting before the natural zero-crossing of the AC waveform.

Yet there is no need to manually turn off the SCR. it will stop conducting naturally at the zero-crossing of the AC voltage.

- ➔ The SCR stops conducting when the current through it drops below the holding current (after initially been triggered and started conducting of course), typically at the zero-crossing of the AC waveform

4. Suppose we put this circuit on the main supply, what precautions should be taken to measure this circuit? Can we just use our oscilloscope?

- Use of high isolation transformer
- We can't strandred oscilloscope probes directly on high voltage circuits. We can use high voltage probes instead or differential probes.
- Ensure proper grounding

- We have to also check your oscilloscope's maximum input voltage rating.

Triac:

1. What is a Diac ?

A DIAC is a bidirectional trigger diode that can conduct current only after its breakdown voltage is exceeded in either direction. It is commonly used to trigger a Triac .

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2. What can you tell me about the maximum power delivered to the load?

3. What happens/could happen if the load is not purely resistive? Elaborate?

When the load is not purely resistive (such as inductive or capacitive), several problems arise in TRIAC phase control circuits:

Inductive load: Current lags behind voltage. This delays the zero crossing point, which can cause the TRIAC to run longer, and have trouble shutting down properly.

Capacitive load: DC voltage This results in premature zero crossing. Capacitor loads can cause high dV/dt , causing the TRIAC to accidentally trigger in the absence of a proper gate pulse. Additionally, the input current of the capacitor components can stress or damage the TRIAC. In both cases The circuit may flicker. Shows inconsistent power distribution. and even components fail. The use of a high rated snubber circuit, filter or TRIAC is recommended to reduce these problems.

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4. Why is it better to control a TRIAC with a pulse and not with a sine wave?

Pulse triggering ensures that the TRIAC is only activated at the desired point in the AC cycle. With a continuous sine wave trigger, energy would be wasted in constantly applying the signal to the gate, which is inefficient and could cause the gate driver circuit to overheat.

Pulse control allows precise triggering of the TRIAC at any point in the AC cycle. A sine wave trigger would not offer the same precision.

Mosfet & IGBT

1. We will use the IGBT with part number NGTB15N60, discuss the most important specifications of this transistor

ABSOLUTE MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Collector-emitter voltage	V_{CES}	600	V
Collector current @ $T_C = 25^\circ\text{C}$ @ $T_C = 100^\circ\text{C}$	I_C	30 15	A
Pulsed collector current, T_{pulse} limited by $T_{J\text{max}}$	I_{CM}	120	A
Diode forward current @ $T_C = 25^\circ\text{C}$ @ $T_C = 100^\circ\text{C}$	I_F	30 15	A
Diode pulsed current, T_{pulse} limited by $T_{J\text{max}}$	I_{FM}	120	A
Gate-emitter voltage	V_{GE}	± 20	V
Power dissipation @ $T_C = 25^\circ\text{C}$ @ $T_C = 100^\circ\text{C}$	P_D	117 47	W
Short circuit withstand time $V_{GE} = 15\text{ V}$, $V_{CE} = 400\text{ V}$, $T_J \leq +150^\circ\text{C}$	t_{SC}	10	μs
Operating junction temperature range	T_J	-55 to +150	$^\circ\text{C}$
Storage temperature range	T_{stg}	-55 to +150	$^\circ\text{C}$
Lead temperature for soldering, 1/8" from case for 5 seconds	T_{SLD}	260	$^\circ\text{C}$

2. We will be using the IGBT as a switch. How much is the safe minimum gate voltage needed to switch the load fully ON?

the safe minimum gate voltage to fully switch the load ON and ensure proper conduction is typically between 10V and 15V.

3. Can you find 3 examples of an inductive load?

Electric motors – Relays – Transformers

4. Can you find an examples of a load where the inductance is parasitic (unwanted)?

Long wires or PCB traces.

5. Will be using the TVS diode: 1.5KE22CA. What is the peak power this diode can handle? What is the clamping voltage? Is this a bi- or unidirectional TVS diode?

Peak Power Handling: 1500W (1.5 kW).

Clamping Voltage: Approximately 35.5V.

Type: **Bidirectional** TVS diode it's indicated by "CA" in the part number.
